

An Efficient Encryption Scheme Based on $(U + V, U + W)$ Codes

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Abstract

In this paper, we propose an improvement to the McEliece encryption scheme by replacing the Goppa code with a $(U+V, U+W)$ code. Specifically, we embed the generator matrices of a split Reed-Solomon code into the generator matrix of the $(U+V, U+W)$ code. This approach disrupts the algebraic structure of Reed-Solomon codes, thereby enhancing resistance against structural attacks targeting such codes, while simultaneously preserving their excellent error-correcting capabilities. As a result, the proposed scheme achieves a significant reduction in public key size. Under the hardness assumptions of the decoding problem and the code distinguishing problem for $(U+V, U+W)$ codes, we prove that the scheme achieves indistinguishability under chosen-plaintext attacks (IND-CPA security). Finally, we provide recommended parameters for various security levels and compare the proposed scheme with other code-based public key encryption schemes.

Keywords: Code-based cryptography, $(U+V, U+W)$ codes, Reed-Solomon code, Public key encryptions

1 Introduction

Code-based public-key cryptosystems are among the earliest public-key cryptographic constructions. In 1978, McEliece [1] proposed the first code-based public-key encryption scheme, relying on the computational hardness of the general decoding problem.

The scheme features a distinctive design: the private key consists of an efficiently decodable Goppa code, while the public key is a structurally obfuscated generator matrix of the same code. This design hides the algebraic structure of the Goppa code, effectively transforming the difficulty of decoding a random linear code into a foundation for cryptographic security. Since its introduction, the McEliece scheme has withstood nearly half a century of intensive cryptanalysis and is still widely regarded as one of the most quantum-resistant public-key encryption schemes available today.

In 1986, Niederreiter proposed a variant of the McEliece encryption scheme, known as the Niederreiter cryptosystem, based on the computational hardness of the syndrome decoding problem. Just as the general decoding problem is equivalent to the syndrome decoding problem, the McEliece and Niederreiter schemes are also equivalent and can be transformed into one another [2]. From an implementation perspective, the McEliece scheme uses the generator matrix of a code as the public key, while the private key contains an efficient decoding algorithm for a structured code. During encryption, the plaintext message is first encoded into a codeword using the generator matrix, and then a low-weight error vector is added. The resulting corrupted codeword serves as the ciphertext. Decryption is performed using the private decoding algorithm to recover the original codeword and extract the plaintext. The Niederreiter scheme, on the other hand, employs a parity-check matrix as the public key, with the private key also consisting of a decoding algorithm for a structured code. Encryption and decryption in this scheme require a publicly known invertible function. Specifically, the plaintext message is first mapped to a low-weight error vector, which is then multiplied by the parity-check matrix to generate the syndrome, serving as the ciphertext. Decryption uses the private decoding algorithm to recover the error vector from the syndrome, and the inverse of the mapping function is then applied to retrieve the original plaintext message.

It is worth noting that both the McEliece and Niederreiter cryptosystems involve large matrix structures in their public keys: the McEliece scheme uses a generator matrix, while the Niederreiter scheme employs a parity-check matrix. The presence of these matrix structures results in significantly larger public key sizes compared to traditional cryptographic systems based on number-theoretic problems. Consequently, considerable research efforts have been devoted to reducing the public key size of code-based cryptosystems. Approaches include the use of structured codes such as Reed-Solomon codes, algebraic geometry codes, and Reed-Muller codes, etc. However, to date, these schemes have been shown to be vulnerable to structural attacks, and many have been completely broken.

In 1992, Sidelnikov et al. [3] successfully broke the Niederreiter cryptosystem based on Reed-Solomon codes by exploiting their algebraic structure. Subsequently, Minder et al. [4, 5] used information set decoding (ISD) techniques to compromise algebraic geometry codes defined over curves of genus at most two. Shortly after, Couvreur et al. [6] demonstrated that algebraic geometry codes constructed from high-genus curves can also be broken in polynomial time. Furthermore, in [7], they leveraged error-correcting pair (ECP) decoding techniques [8, 9] to develop a general attack applicable to algebraic geometry codes defined over curves of arbitrary genus, undermining the security of cryptosystems based on algebraic geometry codes. To enhance the security

of McEliece-type cryptosystems based on generalized Reed-Solomon codes, Baldi et al. [10] proposed a modification that replaces the permutation matrix with a sparse scrambling matrix. However, this approach represents a trade-off between security and error-correction performance. In 2019, Zhang et al. [11] introduced ECC², a scheme that employs elliptic codes and list decoding algorithms to correct more errors than half the minimum distance, thus providing strong resistance against algebraic attacks. Additionally, ECC² exhibits favorable performance in terms of key size and ciphertext expansion.

In 2016, reference [12] introduced a public-key encryption scheme that combines Reed-Solomon codes with $(U, U+V)$ codes. The scheme leverages the special structure of the $(U, U+V)$ code to obscure the structure of the underlying Reed-Solomon codes generator matrix. In the same year, Wang [13] proposed a public-key encryption scheme based on random linear codes, called RLCE, which was submitted to the NIST post-quantum cryptography standardization process and selected as one of the 69 Round-1 candidate schemes [14]. The work in [13] analyzed instances of RLCE constructed using generalized Reed-Solomon codes. By inserting random columns into the generator matrix of a generalized Reed-Solomon code, the scheme disrupts its algebraic structure, thereby rendering classical attacks on generalized Reed-Solomon codes ineffective. However, Couvreur et al. [15] later proposed a polynomial-time key recovery attack against RLCE using a square code distinguisher. Their attack breaks all parameter sets where the number of inserted random columns w is less than $n - k$, effectively rendering these configurations insecure. While parameter sets with more random columns can resist this attack, they result in significantly larger key sizes, which limits the practicality of the scheme.

In 2019, NIST announced the second-round candidates for post-quantum cryptographic standardization. Several of these were code-based cryptosystems, such as BIKE [16], LEDAcrypt [17], and Classic McEliece [18]. However, most of them were found to be potentially vulnerable to key-recovery attacks described in [19, 20]. During the third round [21], BIKE underwent significant modifications: BIKE-1 and BIKE-3 were eliminated, and the parameters of BIKE-2 were adjusted to achieve IND-CCA security. The parameters and key generation of Classic McEliece were also optimized in the third round [22], although its public key size remained the same as in the second round. LEDAcrypt, however, was eliminated in the third round due to its second version is insecure. Ultimately, both BIKE and Classic McEliece advanced to the fourth round of the NIST competition. In 2018, inspired by the Ring-Learning With Errors cryptosystem, Aguilar-Melchor et al. [23] proposed a novel and efficient cryptographic scheme based on the hardness of decoding random quasi-cyclic codes. Unlike previous constructions, this scheme does not conceal the structure of the code in the public matrix, representing a major breakthrough in the design paradigm of code-based encryption. This approach opened a new direction for research in code-based public-key cryptography. In [23], two schemes were proposed: HQC and RQC. HQC is a code-based scheme under the Hamming metric, while RQC is constructed under the rank metric. Both HQC [24] and RQC [25] were submitted to the NIST post-quantum cryptography standardization project. RQC advanced to the second round, while HQC progressed further to the fourth round and was officially announced on

March 11, 2025, as the fifth standardized post-quantum cryptographic algorithm by NIST.

Rank-metric codes are considered one of the most promising candidates for code-based cryptography, as the general decoding problem under the rank metric appears to be significantly harder than its Hamming metric counterpart for the same code length and alphabet size. Consequently, rank-based schemes can achieve smaller key sizes at equivalent security levels. The first rank-based cryptosystem was the Gabidulin-Paramonov-Tretjakov (GPT) scheme [26], a McEliece-type cryptosystem that employs Gabidulin codes. However, GPT and most of its variants [27–29] have been broken by attacks that exploit the specific structure of Gabidulin codes [30–32]. Song et al. [33] investigated the rank decoding problem in conjunction with Low Rank Parity Check (LRPC) codes with block structures. They proposed a secure rank-based cryptosystem whose combined public key and ciphertext sizes are only 50% of those in the original RQC scheme, and significantly smaller than the sizes of code-based submissions in NIST Round 4 such as HQC, BIKE, and Classic McEliece. In 2024, Aragon et al. [34] introduced and analyzed a new rank-metric-based Key Encapsulation Mechanism (KEM) called LowMS. This scheme substantially outperforms all previous Gabidulin-based constructions, particularly in terms of reducing public key size. Its security is based on the hardness of the decisional Rank Support Learning (DRSL) problem and the IND-Gab distinguishing problem.

Our Contributions. In this paper, we propose a new class of McEliece-type public key encryption schemes by integrating $(U+V, U+W)$ codes with Reed-Solomon codes. One of our key contributions is the generalization of the well-known $(U, U+V)$ code construction to $(U+V, U+W)$ codes. We formally prove that, under appropriate parameter settings, $(U+V, U+W)$ codes are computationally indistinguishable from random linear codes of the same length and dimension.

The second contribution of this paper lies in the novel embedding of Reed-Solomon codes into the $(U+V, U+W)$ framework. Specifically, we split the generator matrix of a Reed-Solomon code into two random matrices and embed them into the generator matrix of the $(U+V, U+W)$ code. This design breaks the algebraic structure of the Reed-Solomon code—thereby mitigating structural attacks—while preserving its excellent error-correcting capabilities.

Our third contribution is the introduction of a new construction paradigm for code-based encryption schemes. Unlike traditional approaches that rely on specific families of structured codes, our method allows for the use of any efficiently decodable code, including those previously considered insecure due to their algebraic structures (e.g., algebraic geometry codes). The resulting public key matrices are computationally indistinguishable from random matrices.

Organization. The remaining sections of the paper are organized as follows. Section 2 reviews basic definitions in coding theory and encryption, along with the relevant notations. Section 3 introduces the definition of the $(U+V, U+W)$ code and details the construction of the proposed encryption scheme. In Section 4, we prove that our scheme achieves indistinguishability under chosen-plaintext attacks. Section 5 presents

recommended parameters for various security levels and offers a comparative analysis with existing code-based encryption schemes. Finally, Section 6 concludes the paper.

2 Preliminaries

In this section, some notations and fundamental concepts that will be utilized in the subsequent sections are introduced.

General Notations. Let $\mathbb{F}_q$ denote the finite field of size q . Bold lowercase letters (e.g., $\mathbf{x}$) represent row vectors, and bold uppercase letters (e.g., $\mathbf{H}$) denote matrices. The notation $\mathbf{x}^T$ (resp. $\mathbf{H}^T$) denotes the transpose of the vector $\mathbf{x}$ (resp. matrix $\mathbf{H}$). The concatenation of two vectors $\mathbf{x}$ and $\mathbf{y}$ is denoted by $(\mathbf{x}||\mathbf{y})$, and similarly for matrices $(\mathbf{H}_1||\mathbf{H}_2)$. Let m, n be positive integers such that $n > m$. The notation $[n]$ and $[m; n]$ denote the integer sets $\{1, 2, \dots, n\}$ and $\{m, m+1, \dots, n\}$, respectively. Let S_n denote the set of all permutations on n elements, and let $\mathcal{DP}(n)$ denote the set of all matrices of size $n \times n$ over $\mathbb{F}_q$ that are invertible diagonal-permutation matrices. For a vector $\mathbf{x} \in \mathbb{F}_q^n$, the i -th entry is denoted by $\mathbf{x}(i)$, sometimes written as x_i , so that $\mathbf{x} = (x_i)_{i \in [n]}$. For any subset $\mathcal{I} \subseteq [n]$, $\mathbf{x}_{\mathcal{I}}$ denotes the vector whose coordinates are indexed by $\mathcal{I}$, i.e., $\mathbf{x}_{\mathcal{I}} = (x_i)_{i \in \mathcal{I}}$ where $\mathbf{x} = (x_i)_{i \in [n]}$. The Hamming weight of $\mathbf{x} \in \mathbb{F}_q^n$, denoted by $|\mathbf{x}|$ or $wt(\mathbf{x})$, is defined as the number of nonzero entries in $\mathbf{x}$, i.e.,

$$|\mathbf{x}| = |\{i \in [n] \mid \mathbf{x}(i) \neq 0\}|.$$

Let S be a finite set. The notation $s \stackrel{\$}{\leftarrow} S$ denotes that the element s is sampled uniformly at random from the set S . For convenience, $|S|$ or $\#S$ denotes the size of the set S . Let $\mathcal{E}_{q,n}^w$ denote the set of vectors in $\mathbb{F}_q^n$ having Hamming weight exactly w . The size of this set is $\binom{n}{w}(q-1)^w$. Let X be a random variable. Then $\mathbb{P}(X = A)$ denotes the probability that event A occurs, and $\mathbb{E}(X)$ denotes the expected value of X . Let $\mathcal{D}_1$ and $\mathcal{D}_2$ be two discrete probability distributions over the same probability space E . The statistical distance between them is defined as

$$\rho(\mathcal{D}_1, \mathcal{D}_2) = \frac{1}{2} \sum_{x \in E} |\mathcal{D}_1(x) - \mathcal{D}_2(x)|.$$

Coding Theory. An $[n, k]$ linear code $\mathcal{C}$ over a finite field $\mathbb{F}_q$ is defined as a k -dimensional subspace of the n -dimensional vector space $\mathbb{F}_q^n$. The elements of $\mathcal{C}$ are called codewords, where n is the length of the code and k is its dimension. Analogous to representing a vector space using a basis, a linear code $\mathcal{C}$ can also be represented by a set of basis codewords, which form a matrix known as the generator matrix of $\mathcal{C}$.

Definition 1. Let $\mathcal{C}$ be an $[n, k]$ linear code over $\mathbb{F}_q$. A generator matrix of $\mathcal{C}$ is a full-rank $k \times n$ matrix over $\mathbb{F}_q$ whose rows form a basis of $\mathcal{C}$. If

$$\mathcal{C} = \{\mathbf{x}\mathbf{G} \mid \mathbf{x} \in \mathbb{F}_q^k\},$$

then $\mathbf{G}$ is a generator matrix of $\mathcal{C}$.

A parity-check matrix of $\mathcal{C}$ is a full-rank $(n - k) \times n$ matrix over $\mathbb{F}_q$. If

$$\forall \mathbf{x} \in \mathcal{C} \Leftrightarrow \mathbf{x}\mathbf{H}^T = \mathbf{0},$$

then $\mathbf{H}$ is a parity-check matrix of $\mathcal{C}$.

Note that both the generator matrix and the parity-check matrix of a linear code $\mathcal{C}$ are not unique. For any generator matrix $\mathbf{G}$ and parity-check matrix $\mathbf{H}$ of $\mathcal{C}$, it always holds that $\mathbf{G}\mathbf{H}^T = \mathbf{0}$. An information set of an $[n, k]$ linear code $\mathcal{C}$ is a subset $I \subset [n]$ of k coordinate positions corresponding to linearly independent columns of the generator matrix $\mathbf{G}$. Its complement corresponds to $r = n - k$ linearly independent columns in the parity-check matrix $\mathbf{H}$. Given a parity-check matrix $\mathbf{H} \in \mathbb{F}_q^{r \times n}$ of $\mathcal{C}$ and an information set I , for any $\mathbf{s} \in \mathbb{F}_q^r$ and $\mathbf{x} \in \mathbb{F}_q^n$, there exists a unique vector $\mathbf{e} \in \mathbb{F}_q^n$ such that $\mathbf{e}\mathbf{H}^T = \mathbf{s}$ and $\mathbf{e}_I = \mathbf{x}_I$.

Definition 2. Let $\mathcal{C}$ be an $[n, k]$ linear code over the finite field $\mathbb{F}_q$. The dual code of $\mathcal{C}$, denoted by $\mathcal{C}^\perp$, is defined as:

$$\mathcal{C}^\perp = \{\mathbf{y} \in \mathbb{F}_q^n \mid \forall \mathbf{x} \in \mathcal{C}, \langle \mathbf{x}, \mathbf{y} \rangle = 0\},$$

where $\langle \mathbf{x}, \mathbf{y} \rangle$ denotes the standard inner product of vectors $\mathbf{x}$ and $\mathbf{y}$. The dual code $\mathcal{C}^\perp$ is an $[n, r]$ linear code, where $r = n - k$. Its generator matrix and parity-check matrix are, respectively, the parity-check matrix and generator matrix of the original code $\mathcal{C}$. If $\mathcal{C} = \mathcal{C}^\perp$, then $\mathcal{C}$ is called a self-dual code.

Definition 3. Let $\vec{\alpha} = (\alpha_1, \alpha_2, \dots, \alpha_n)$ be a vector of distinct nonzero elements in the finite field $\mathbb{F}_q$. The k -dimensional Reed-Solomon code $\mathcal{C}$ defined by $\vec{\alpha}$ is given

$$\mathcal{C} = \{(f(\alpha_1), f(\alpha_2), \dots, f(\alpha_n)) \mid \deg(f) < k\}.$$

Let $f_j(x) = x^j$ for $j = 0, \dots, k - 1$. Then the matrix

$$\mathbf{G} = \begin{pmatrix} f_0(\alpha_1) & f_0(\alpha_2) & \cdots & f_0(\alpha_n) \\ f_1(\alpha_1) & f_1(\alpha_2) & \cdots & f_1(\alpha_n) \\ \vdots & \vdots & \ddots & \vdots \\ f_{k-1}(\alpha_1) & f_{k-1}(\alpha_2) & \cdots & f_{k-1}(\alpha_n) \end{pmatrix} = \begin{pmatrix} 1 & 1 & \cdots & 1 \\ \alpha_1 & \alpha_2 & \cdots & \alpha_n \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_1^{k-1} & \alpha_2^{k-1} & \cdots & \alpha_n^{k-1} \end{pmatrix},$$

is a generator matrix of the code $\mathcal{C}$. The dual code $\mathcal{C}^\perp$ is itself an $[n, n - k]$ Reed-Solomon code.

The General Decoding (GD) problem is one of the most fundamental and widely used hardness assumptions in code-based cryptographic schemes. The security of our proposed UVW scheme is also based on the intractability of this problem.

Problem 1. Given a generator matrix $\mathbf{G} \in \mathbb{F}_q^{k \times n}$ for an $[n, k]$ linear code $\mathcal{C}$ over a finite field $\mathbb{F}_q$, a vector $\mathbf{r} \in \mathbb{F}_q^n$, and a positive integer w , the objective is to find a message vector $\mathbf{m} \in \mathbb{F}_q^k$ such that $\text{wt}(\mathbf{m}\mathbf{G} - \mathbf{r}) \leq w$.

Public Key Encryption. We now review the definitions of the public key encryption scheme and the security model.

Definition 4. A typical public-key encryption scheme consists of the following three polynomial time algorithms: KeyGen , Enc , and Dec .

- $\text{KeyGen}(1^\lambda)$: This is a probabilistic polynomial time algorithm that, given a security parameter 1^λ , outputs a pair of keys (pk, sk) . We denote this as $(\text{pk}, \text{sk}) \leftarrow \text{KeyGen}(1^\lambda)$, where pk is the public key, which is made available to everyone for encrypting messages, and sk is the secret key, which is kept private and used for decryption;
- $\text{Enc}(\text{pk}, \mathbf{m})$: This is a probabilistic polynomial time algorithm that takes a message $\mathbf{m}$ and the public key pk as input and outputs a ciphertext $\mathbf{c}$. We denote this as $\mathbf{c} \leftarrow \text{Enc}(\text{pk}, \mathbf{m})$;
- $\text{Dec}(\text{sk}, \mathbf{c})$: This is a deterministic polynomial time algorithm that takes a ciphertext $\mathbf{c}$ and the secret key sk as input, and returns the original message $\mathbf{m}$. If the ciphertext is invalid, the algorithm returns a special symbol $\perp$ to indicate decryption failure. We denote this as $\mathbf{m} = \text{Dec}(\text{sk}, \mathbf{c})$.

In some cases, an additional initialization step, $\text{Setup}(1^\lambda)$, is introduced as part of the scheme description. This algorithm takes the system security parameter λ as input and outputs a set of public parameters params . The key generation algorithm is then invoked as $\text{KeyGen}(\text{params})$ to produce the public and private key pair. The encryption and decryption algorithms remain unchanged thereafter.

Correctness. For every key pair $(\text{pk}, \text{sk}) \leftarrow \text{KeyGen}(1^\lambda)$ and any message $\mathbf{m}$, the following correctness property must hold:

$$\text{Dec}(\text{sk}, \text{Enc}(\text{pk}, \mathbf{m})) = \mathbf{m}. \quad (1)$$

This guarantees that, given a valid key pair, any message encrypted using the public key can be correctly and faithfully recovered using the corresponding secret key. In other words, for any message $\mathbf{m}$, encrypting it to obtain a ciphertext $\mathbf{c}$ and then decrypting $\mathbf{c}$ must yield the original message $\mathbf{m}$.

Indistinguishability under Chosen-Plaintext Attack (IND-CPA) and Indistinguishability under Chosen-Ciphertext Attack (IND-CCA) are two foundational security notions for public-key encryption schemes. In the IND-CPA model, the adversary is allowed to obtain ciphertexts for arbitrary plaintexts and aims to distinguish which of two chosen messages corresponds to a given challenge ciphertext.

Let $\Pi = (\text{KeyGen}, \text{Enc}, \text{Dec})$ be a public-key encryption scheme. The formal definition of IND-CPA security is given below:

Definition 5. Let $\Pi = (\text{KeyGen}, \text{Enc}, \text{Dec})$ be a public-key encryption scheme, and let λ be the security parameter. We say that Π is IND-CPA secure if, for every probabilistic polynomial time (PPT) adversary $\mathcal{A}$, the probability of experiment $\text{Exp}_{\mathcal{A}, \Pi}^{\text{IND-CPA}}(1^\lambda)$ outputting 1 is close to $1/2$, that is:

$$\left| \mathbb{P}[\text{Exp}_{\mathcal{A}, \Pi}^{\text{IND-CPA}}(1^\lambda) = 1] - \frac{1}{2} \right| \leq \text{negl}(\lambda).$$

$\text{Exp}_{\mathcal{A}, \Pi}^{\text{IND-CPA}}(1^\lambda) :$

- 1) The challenger generates a key pair: $(\text{pk}, \text{sk}) \leftarrow \text{KeyGen}(1^\lambda)$;
- 2) The adversary $\mathcal{A}$ outputs two messages: $(\mathbf{m}_0, \mathbf{m}_1) \leftarrow \mathcal{A}(\text{pk}, 1^\lambda)$ (Note: The adversary is given the public key and may access an encryption oracle, therefore we omit the encryption queries here.);
- 3) $b \xleftarrow{\$} \{0, 1\}$. The challenger computes the challenge ciphertext $\mathbf{c} \leftarrow \text{Enc}(\text{pk}, \mathbf{m}_b)$ and sends it to $\mathcal{A}$;
- 4) $b' \leftarrow \mathcal{A}(\text{pk}, 1^\lambda, \mathbf{c})$;
- 5) The experiment outputs 1 if $b' = b$, and 0 otherwise.

Compared to IND-CPA security, a stronger notion is indistinguishability under chosen-ciphertext attacks (IND-CCA), where the adversary is additionally granted access to a decryption oracle. For a public-key encryption scheme that is already IND-CPA secure, it is often possible to achieve IND-CCA security through generic transformations such as the Naor-Yung transformation [35] or the Fujisaki-Okamoto transformation [36].

3 Code-Based Encryption Scheme

In this section, we first present the definition of the (generalized) $(U + V, U + W)$ codes. We then provide a detailed description of two public-key encryption schemes, UVW1 and UVW2, constructed based on these codes. Specifically, UVW1 is designed using the $(U + V, U + W)$ codes, while UVW2 is constructed from the generalized $(U + V, U + W)$ codes.

3.1 $(U + V, U + W)$ Codes

The $(U + V, U + W)$ codes is a generalization of the $(U, U + V)$ codes. Its formal definition is given below.

Definition 6 ($(U + V, U + W)$ Codes). A $(U + V, U + W)$ code over the finite field $\mathbb{F}_q$ is defined by three linear codes U, V , and W . Let n, k_1 , and k_2 be three positive integers, where n is even. Suppose U is an $[\frac{n}{2}, k_1]$ linear code, and both V and W are $[\frac{n}{2}, k_2]$ linear codes. Then, the generator matrix of the $(U + V, U + W)$ code is defined as follows:

$$\mathbf{G} = \begin{pmatrix} \mathbf{G}_U & \mathbf{G}_U \\ \mathbf{G}_V & \mathbf{G}_W \end{pmatrix},$$

where $\mathbf{G}_U, \mathbf{G}_V$, and $\mathbf{G}_W$ are the generator matrices of U, V , and W , respectively.

The generalized $(U + V, U + W)$ codes is a further extension of the $(U + V, U + W)$ codes. Its formal definition is provided in Definition 7.

Definition 7 (Generalized $(U + V, U + W)$ Codes). A generalized $(U + V, U + W)$ code over the finite field $\mathbb{F}_q$ is defined using three linear codes U, V , and W . Let n, k_1 , and k_2 be three positive integers, where n is even. Suppose U is an $[\frac{n}{2}, k_1]$ linear code, and both V and W are $[\frac{n}{2}, k_2]$ linear codes. Let $\mathbf{D} \in \mathbb{F}_q^{n \times n}$ be a randomly chosen

invertible diagonal permutation matrix. Then, the generator matrix of the generalized $(U + V, U + W)$ code is defined as follows:

$$\mathbf{G} = \begin{pmatrix} \mathbf{G}_U & \mathbf{G}_V \\ \mathbf{G}_V & \mathbf{G}_W \end{pmatrix} \mathbf{D},$$

where $\mathbf{G}_U, \mathbf{G}_V$, and $\mathbf{G}_W$ are the generator matrices of U, V , and W , respectively.

Proposition 1. Let $\mathcal{C}$ be a generalized $[n, k]$ $(U + V, U + W)$ code over the finite field $\mathbb{F}_q$. Then its dual code is given by

$$\mathcal{C}^\perp = \{\mathbf{y} \in \mathbb{F}_q^n \mid \forall \mathbf{x} \in \mathcal{C}, \langle \mathbf{x}, \mathbf{y} \rangle = 0\},$$

and $\mathcal{C}^\perp$ is also a generalized $[n, n - k]$ $(U + V, U + W)$ code.

Proposition 1 can be proven straightforwardly. By Definition 2, the parity-check matrix of code $\mathcal{C}$ is exactly the generator matrix of its dual code $\mathcal{C}^\perp$. Let

$$\mathbf{G} = \begin{pmatrix} \mathbf{G}_U & \mathbf{G}_V \\ \mathbf{G}_V & \mathbf{G}_W \end{pmatrix} \mathbf{D}$$

be the generator matrix of code $\mathcal{C}$, where $\mathbf{G}_U \in \mathbb{F}_q^{k_1 \times n/2}, \mathbf{G}_V \in \mathbb{F}_q^{k_2 \times n/2}, \mathbf{G}_W \in \mathbb{F}_q^{k_2 \times n/2}, k = k_1 + k_2$. Let

$$\mathbf{H} = \begin{pmatrix} \mathbf{H}_X & \mathbf{H}_Y \\ \mathbf{H}_Z & \mathbf{H}_Z \end{pmatrix} \begin{pmatrix} -\mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{pmatrix} \mathbf{D}^{-1} \quad (2)$$

be the generator matrix of dual code $\mathcal{C}^\perp$. If $\mathbf{H}_X, \mathbf{H}_Y \in \mathbb{F}_q^{(n-k) \times n/2}, \mathbf{H}_X, \mathbf{H}_Y \in \mathbb{F}_q^{(n-k) \times n/2}$ and $\mathbf{H}_Z$ does not exist, then $\mathcal{C}^\perp$ is not a $(U + V, U + W)$ code. Otherwise, if $\mathbf{H}_Z$ exists, then $\mathcal{C}^\perp$ is a $(U + V, U + W)$ code. Let

$$\mathbf{G}\mathbf{H}^\top = \begin{pmatrix} \mathbf{G}_U & \mathbf{G}_V \\ \mathbf{G}_V & \mathbf{G}_W \end{pmatrix} \begin{pmatrix} -\mathbf{H}_X & \mathbf{H}_Y \\ -\mathbf{H}_Z & \mathbf{H}_Z \end{pmatrix}^\top = \mathbf{0},$$

which yields the following set of equations:

$$\begin{aligned} \mathbf{G}_U(-\mathbf{H}_X + \mathbf{H}_Y)^\top &= \mathbf{0}, \\ -\mathbf{G}_V\mathbf{H}_X^\top + \mathbf{G}_W\mathbf{H}_Y^\top &= \mathbf{0}, \\ (-\mathbf{G}_V + \mathbf{G}_W)\mathbf{H}_Z^\top &= \mathbf{0}. \end{aligned}$$

Given that $\mathbf{G}_V, \mathbf{G}_W \in \mathbb{F}_q^{k_2 \times n/2}$, it follows that there exists a generator matrix $\mathbf{H}$ for $\mathcal{C}^\perp$ as defined in equation (2), where $\mathbf{H}_Z \in \mathbb{F}_q^{(n/2-k_2) \times n/2}$. Therefore, the dual code $\mathcal{C}^\perp$ is also a $(U + V, U + W)$ code.

3.2 The UVW1 Public-Key Encryption Scheme

The UVW1 public-key encryption scheme consists of the following four algorithms: Setup, KeyGen, Enc, and Dec.

Algorithm 1 Setup(1^λ)

Require: security parameter 1^λ

Ensure: system parameters **params**

- 1: Choose appropriate system parameters based on the security parameter, denoted as **params** = $(q, n, k, k_1, k_2, w, m)$, where $k = k_1 + k_2$;
 - 2: Select a secure hash function $\mathcal{H} : \{0, 1\}^* \rightarrow \mathbb{F}_2^m$;
 - 3: Define the finite field $\mathbb{F}_q$, over which all subsequent operations are performed;
 - 4: **return** **params**
-

Algorithm 2 KeyGen(**params**)

Require: system parameters **params**

Ensure: (pk, sk)

- 1: Randomly choose two full-rank matrices $\mathbf{G}_U \in \mathbb{F}_q^{k_1 \times n/2}$ and $\mathbf{G}_W \in \mathbb{F}_q^{k_2 \times n/2}$, a nonsingular matrix $\mathbf{S} \in \mathbb{F}_q^{k \times k}$, and a permutation matrix $\mathbf{P} \in \mathbb{F}_q^{n \times n}$;
 - 2: Randomly choose a $[n/2, k_2]$ Reed-Solomon code $\mathcal{C}$ over the finite field $\mathbb{F}_q$ that can correct up to w errors, where $w \leq n/2 - \sqrt{nk_2/2}$. Let $\mathbf{G}_{RS} \in \mathbb{F}_q^{k_2 \times n/2}$ be a generator matrix of $\mathcal{C}$;
 - 3: Let $\mathbf{G}_V = \mathbf{G}_{RS} + \mathbf{G}_W$, $\mathbf{G}_{sk} = \begin{pmatrix} \mathbf{G}_U & \mathbf{G}_U \\ \mathbf{G}_V & \mathbf{G}_W \end{pmatrix}$, $\mathbf{G}_{pk} = \mathbf{S}\mathbf{G}_{sk}\mathbf{P}$;
 - 4: Let sk = ($\mathbf{S}, \mathbf{G}_{sk}, \mathbf{P}$), and pk = $\mathbf{G}_{pk}$;
 - 5: **return** (pk, sk)
-

Algorithm 3 Enc(pk, m)

Require: public key pk and the message $\mathbf{m} \in \mathbb{F}_2^m$

Ensure: ciphertext **c**

- 1: Randomly select a vector $\mathbf{r} \xleftarrow{\$} \mathbb{F}_q^k$;
 - 2: Randomly choose a vector $\mathbf{e} \in \mathcal{E}_{q,n}^w$ of Hamming weight w , where all nonzero components are distinct;
 - 3: Compute $\mathbf{c}_1 = \mathbf{r}\mathbf{G}_{pk} + \mathbf{e}$ and $\mathbf{c}_2 = \mathbf{m} \oplus \mathcal{H}(\mathbf{r}, \mathbf{e})$;
 - 4: **return** $\mathbf{c} = (\mathbf{c}_1, \mathbf{c}_2)$
-

Algorithm 4 Dec(sk, c)

Require: secret key sk and the ciphertext $\mathbf{c} = (\mathbf{c}_1, \mathbf{c}_2)$

Ensure: message **m**

- 1: Compute $\mathbf{c}_1\mathbf{P}^{-1} = (\mathbf{c}_{11} \parallel \mathbf{c}_{12})$. Use the Reed-Solomon decoding algorithm to find a vector $\mathbf{r}'_2$ and an error vector $\bar{\mathbf{e}}$ such that $\mathbf{r}'_2\mathbf{G}_{RS} + \bar{\mathbf{e}} = \mathbf{c}_{11} - \mathbf{c}_{12}$;
 - 2: Compute $\mathbf{r}'_1 = (\mathbf{c}_{12} - \mathbf{r}'_2\mathbf{G}_W)_{I_1}(\mathbf{G}_U)_{I_1}^{-1}$, where $I_1 \subseteq I = [n/2] \setminus \text{supp}(\bar{\mathbf{e}})$ and $|I_1| = k_1$;
 - 3: Compute $\mathbf{r}' = (\mathbf{r}'_1 \parallel \mathbf{r}'_2)\mathbf{S}^{-1}$, and let $\mathbf{e}' = \mathbf{c}_1 - \mathbf{r}'\mathbf{G}_{pk}$;
 - 4: **return** $\mathbf{m}' = \mathbf{c}_2 \oplus \mathcal{H}(\mathbf{r}', \mathbf{e}')$.
-

Correctness. Given that $\mathbf{r}\mathbf{S}\mathbf{G}_{sk}\mathbf{P} + \mathbf{e} = \mathbf{c}_1$, let

$$\mathbf{r}\mathbf{S} = (\mathbf{r}_1 \parallel \mathbf{r}_2), \quad \mathbf{e}\mathbf{P}^{-1} = (\mathbf{e}_1 \parallel \mathbf{e}_2), \quad \mathbf{c}_1\mathbf{P}^{-1} = (\mathbf{c}_{11} \parallel \mathbf{c}_{12}).$$

Then, according to the definition of the secret key matrix $\mathbf{G}_{\text{sk}}$, we have:

$$\mathbf{r}_1 \mathbf{G}_U + \mathbf{r}_2 (\mathbf{G}_{RS} + \mathbf{G}_W) + \mathbf{e}_1 = \mathbf{c}_{11}, \quad \mathbf{r}_1 \mathbf{G}_U + \mathbf{r}_2 \mathbf{G}_W + \mathbf{e}_2 = \mathbf{c}_{12}. \quad (3)$$

Subtracting the second equation from the first yields:

$$\mathbf{r}_2 \mathbf{G}_{RS} + \mathbf{e}_1 - \mathbf{e}_2 = \mathbf{c}_{11} - \mathbf{c}_{12},$$

where the vectors $\mathbf{r}_2$ and $\bar{\mathbf{e}} = \mathbf{e}_1 - \mathbf{e}_2$ are exactly the values recovered in the first step of the decryption algorithm. After obtaining the vectors $\mathbf{r}_2$ and $\mathbf{e}_1 - \mathbf{e}_2$, we use the second equation in (3) to derive

$$\mathbf{r}_1 \mathbf{G}_U + \mathbf{e}_2 = \mathbf{c}_{12} - \mathbf{r}_2 \mathbf{G}_W.$$

Since each non-zero component of the error vector $\mathbf{e}$ is distinct, it follows that $\text{supp}(\mathbf{e}_2) \subseteq \text{supp}(\bar{\mathbf{e}})$. Thus, for any $i \in I = [n/2] \setminus \text{supp}(\bar{\mathbf{e}})$, we have $\mathbf{e}_2(i) = 0$, meaning that no errors occurred at these positions. Therefore, selecting any subset $I_1 \subseteq I$ of size k_1 , we have

$$\mathbf{r}_1 (\mathbf{G}_U)_{I_1} + (\mathbf{e}_2)_{I_1} = (\mathbf{c}_{12} - \mathbf{r}_2 \mathbf{G}_W)_{I_1},$$

which simplifies to

$$\mathbf{r}_1 = (\mathbf{c}_{12} - \mathbf{r}_2 \mathbf{G}_W)_{I_1} (\mathbf{G}_U)_{I_1}^{-1}.$$

Once the vectors $\mathbf{r}_1$ and $\mathbf{r}_2$ are obtained, the original vectors $\mathbf{r}$ and $\mathbf{e}$ can be easily recovered. Hence, the decryption algorithm can successfully recover the message:

$$\text{Dec}(\text{sk}, \text{Enc}(\text{pk}, \mathbf{m})) = \mathbf{m}. \quad (4)$$

It is worth noting that, since the scheme operates over a sufficiently large finite field $\mathbb{F}_q$, the matrix $(\mathbf{G}_U)_{I_1}$ is invertible with overwhelming probability when I_1 is chosen randomly. In the rare case that it is not invertible, one can simply select another subset I_1 . Therefore, the decryption process remains highly efficient.

3.3 The UVW2 Public-Key Encryption Scheme

The UVW2 public-key encryption scheme consists of the following four algorithms: Setup, KeyGen, Enc, and Dec.

Algorithm 5 Setup(1^λ)

Require: security parameter 1^λ

Ensure: system parameters **params**

- 1: Choose appropriate system parameters based on the security parameter, denoted as **params** = $(q, n, k, k_1, k_2, w, m)$, where $k = k_1 + k_2$;
 - 2: Select a secure hash function $\mathcal{H} : \{0, 1\}^* \rightarrow \mathbb{F}_2^m$;
 - 3: Define the finite field $\mathbb{F}_q$, over which all subsequent operations are performed;
 - 4: **return** **params**
-

Algorithm 6 KeyGen(params)

Require: system parameters **params****Ensure:** (pk, sk)

- 1: Randomly choose two full-rank matrices $\mathbf{G}_U \in \mathbb{F}_q^{k_1 \times n/2}$ and $\mathbf{G}_W \in \mathbb{F}_q^{k_2 \times n/2}$, a nonsingular matrix $\mathbf{S} \in \mathbb{F}_q^{k \times k}$, and an invertible diagonal permutation matrix $\mathbf{D} \in \mathbb{F}_q^{n \times n}$;
 - 2: Randomly choose a $[n/2, k_2]$ Reed-Solomon code $\mathcal{C}$ over the finite field $\mathbb{F}_q$ that can correct up to w errors, where $w \leq n/2 - \sqrt{nk_2/2}$. Let $\mathbf{G}_{RS} \in \mathbb{F}_q^{k_2 \times n/2}$ be a generator matrix of $\mathcal{C}$;
 - 3: Let $\mathbf{G}_V = \mathbf{G}_{RS} + \mathbf{G}_W$, $\mathbf{G}_{sk} = \begin{pmatrix} \mathbf{G}_U & \mathbf{G}_U \\ \mathbf{G}_V & \mathbf{G}_W \end{pmatrix}$, $\mathbf{G}_{pk} = \mathbf{S}\mathbf{G}_{sk}\mathbf{D}$;
 - 4: Let $\mathbf{sk} = (\mathbf{S}, \mathbf{G}_{sk}, \mathbf{D})$, and $\mathbf{pk} = \mathbf{G}_{pk}$;
 - 5: **return** (pk, sk)
-

Algorithm 7 Enc(pk, m)

Require: public key **pk** and the message $\mathbf{m} \in \mathbb{F}_2^m$ **Ensure:** ciphertext **c**

- 1: Randomly choose a vector $\mathbf{r} \xleftarrow{\$} \mathbb{F}_q^k$;
 - 2: Randomly choose a vector $\mathbf{e} \in \mathcal{E}_{q,n}^w$;
 - 3: Compute $\mathbf{c}_1 = \mathbf{r}\mathbf{G}_{pk} + \mathbf{e}$ and $\mathbf{c}_2 = \mathbf{m} \oplus \mathcal{H}(\mathbf{r}, \mathbf{e})$;
 - 4: **return** $\mathbf{c} = (\mathbf{c}_1, \mathbf{c}_2)$
-

Algorithm 8 Dec(sk, c)

Require: secret key **sk** and the ciphertext $\mathbf{c} = (\mathbf{c}_1, \mathbf{c}_2)$ **Ensure:** message **m**

- 1: Compute $\mathbf{c}_1\mathbf{D}^{-1} = (\mathbf{c}_{11} \parallel \mathbf{c}_{12})$. Use the Reed-Solomon decoding algorithm to find a vector $\mathbf{r}'_2$ and an error vector $\bar{\mathbf{e}}$ such that $\mathbf{r}'_2\mathbf{G}_{RS} + \bar{\mathbf{e}} = \mathbf{c}_{11} - \mathbf{c}_{12}$;
 - 2: Compute $\mathbf{r}'_1 = (\mathbf{c}_{12} - \mathbf{r}'_2\mathbf{G}_W)_{I_1}(\mathbf{G}_U)_{I_1}^{-1}$, where $I_1 \subseteq I = [n/2] \setminus \text{supp}(\bar{\mathbf{e}})$ and $|I_1| = k_1$;
 - 3: Compute $\mathbf{r}' = (\mathbf{r}'_1 \parallel \mathbf{r}'_2)\mathbf{S}^{-1}$, and let $\mathbf{e}' = \mathbf{c}_1 - \mathbf{r}'\mathbf{G}_{pk}$. If $\text{wt}(\mathbf{e}') = w$, proceed to the next step. Otherwise, return to Step 2 and choose a new subset I_1 to recompute $\mathbf{r}'_1$.
 - 4: **return** $\mathbf{m}' = \mathbf{c}_2 \oplus \mathcal{H}(\mathbf{r}', \mathbf{e}')$.
-

Correctness. Given that $\mathbf{r}\mathbf{S}\mathbf{G}_{sk}\mathbf{D} + \mathbf{e} = \mathbf{c}_1$, let

$$\mathbf{r}\mathbf{S} = (\mathbf{r}_1 \parallel \mathbf{r}_2), \quad \mathbf{e}\mathbf{D}^{-1} = (\mathbf{e}_1 \parallel \mathbf{e}_2), \quad \mathbf{c}_1\mathbf{D}^{-1} = (\mathbf{c}_{11} \parallel \mathbf{c}_{12}).$$

Then, according to the definition of the secret key matrix $\mathbf{G}_{sk}$, we have:

$$\mathbf{r}_1\mathbf{G}_U + \mathbf{r}_2(\mathbf{G}_{RS} + \mathbf{G}_W) + \mathbf{e}_1 = \mathbf{c}_{11}, \quad \mathbf{r}_1\mathbf{G}_U + \mathbf{r}_2\mathbf{G}_W + \mathbf{e}_2 = \mathbf{c}_{12}. \quad (5)$$

Subtracting the second equation from the first yields:

$$\mathbf{r}_2\mathbf{G}_{RS} + \mathbf{e}_1 - \mathbf{e}_2 = \mathbf{c}_{11} - \mathbf{c}_{12},$$

where the vectors $\mathbf{r}_2$ and $\bar{\mathbf{e}} = \mathbf{e}_1 - \mathbf{e}_2$ are exactly the values recovered in the first step of the decryption algorithm. After obtaining the vectors $\mathbf{r}_2$ and $\mathbf{e}_1 - \mathbf{e}_2$, we use the second equation in (5) to derive

$$\mathbf{r}_1\mathbf{G}_U + \mathbf{e}_2 = \mathbf{c}_{12} - \mathbf{r}_2\mathbf{G}_W.$$

Given that $(\mathbf{e}_1 \| \mathbf{e}_2) = \mathbf{e} \mathbf{D}^{-1}$, let $I_2 = \text{supp}(\mathbf{e}_2)$. If, for every $i \in I_2$, it holds that $\mathbf{e}_2(i) \neq \mathbf{e}_1(i)$, then we must have the inclusion

$$\text{supp}(\mathbf{e}_2) \subseteq \text{supp}(\bar{\mathbf{e}}),$$

where $\bar{\mathbf{e}} = \mathbf{e}_1 - \mathbf{e}_2$.

From this inclusion, we can deduce that the entries of $\mathbf{e}_2$ indexed by the set

$$I = [n/2] \setminus \text{supp}(\bar{\mathbf{e}})$$

must be zero. That is, for all $i \in I$, we have $\mathbf{e}_2(i) = 0$.

Therefore, any subset $I_1 \subseteq I$ of size k_1 can be used to compute $\mathbf{r}_1$, since the corresponding coordinates of $\mathbf{e}_2$ are guaranteed to be error-free. Specifically, the following equation holds:

$$\mathbf{r}_1(\mathbf{G}_U)_{I_1} + (\mathbf{e}_2)_{I_1} = (\mathbf{c}_{12} - \mathbf{r}_2 \mathbf{G}_W)_{I_1},$$

which implies:

$$\mathbf{r}_1 = (\mathbf{c}_{12} - \mathbf{r}_2 \mathbf{G}_W)_{I_1} \cdot (\mathbf{G}_U)_{I_1}^{-1}.$$

Once the vectors $\mathbf{r}_1$ and $\mathbf{r}_2$ are obtained, the original vectors $\mathbf{r}$ and $\mathbf{e}$ can be easily recovered. Hence, the decryption algorithm can successfully recover the message:

$$\text{Dec}(\text{sk}, \text{Enc}(\text{pk}, \mathbf{m})) = \mathbf{m}. \quad (6)$$

If there exists an index $i \in I_2$ such that $\mathbf{e}_2(i) = \mathbf{e}_1(i)$, then the inclusion $\text{supp}(\mathbf{e}_2) \subseteq \text{supp}(\bar{\mathbf{e}})$ no longer holds. Consequently, there is a non-negligible probability that the subset $I_1 \subseteq I = [n/2] \setminus \text{supp}(\bar{\mathbf{e}})$ chosen in Step 2 of the decryption algorithm contains some indices where $\mathbf{e}_2$ has nonzero entries. That is,

$$I_1 \cap \text{supp}(\mathbf{e}_2) \neq \emptyset.$$

In such a case, the recovered vector $\mathbf{e}'$ computed in Step 3 will fail to satisfy the condition $wt(\mathbf{e}') = w$, implying that $\mathbf{e}' \neq \mathbf{e}$.

To address this, the decryption algorithm returns to Step 2 and selects a new subset $I_1 \subseteq I = [n/2] \setminus \text{supp}(\bar{\mathbf{e}})$, repeating the computation until a subset satisfying $I_1 \cap \text{supp}(\mathbf{e}_2) = \emptyset$ is found. In this case, we have:

$$\mathbf{r}_1(\mathbf{G}_U)_{I_1} + (\mathbf{e}_2)_{I_1} = (\mathbf{c}_{12} - \mathbf{r}_2 \mathbf{G}_W)_{I_1}, \text{ with } \mathbf{e}_2(i) = 0 \text{ for all } i \in I_1,$$

and thus:

$$\mathbf{r}_1 = (\mathbf{c}_{12} - \mathbf{r}_2 \mathbf{G}_W)_{I_1} \cdot (\mathbf{G}_U)_{I_1}^{-1}.$$

Once the vectors $\mathbf{r}_1$ and $\mathbf{r}_2$ are correctly recovered, the full vectors $\mathbf{r}$ and $\mathbf{e}$ can be accurately reconstructed. Therefore, the following correctness property holds:

$$\text{Dec}(\text{sk}, \text{Enc}(\text{pk}, \mathbf{m})) = \mathbf{m}. \quad (7)$$

The UVW2 public key encryption scheme is similar to UVW1. In the key generation algorithm, UVW1 adopts a construction based on $(U + V, U + W)$, whereas UVW2 employs a generalized $(U + V, U + W)$ construction. Intuitively, UVW2 allows for a broader choice of variables, which results in a greater variety of possible public key matrices. This increased diversity makes it harder to distinguish the public key matrix from a truly random one.

In the encryption algorithm, UVW1 selects an error vector $\mathbf{e} \in \mathcal{E}_{q,n}^w$ of Hamming weight w , where all non-zero entries are distinct. By contrast, UVW2 selects $\mathbf{e} \in \mathcal{E}_{q,n}^w$, which is any vector of Hamming weight w regardless of whether the non-zero entries are distinct. As a result, the ciphertext space of UVW2 is strictly larger than that of UVW1. The ratio of their sizes is given by:

$$\frac{(q-1)^w}{\prod_{i=1}^w (q-i)}. \quad (8)$$

UVW2 not only enlarges the ciphertext space compared to UVW1, but also indirectly mitigates known-information attacks (although such known information does not currently pose a clear threat to the security of the scheme). Naturally, the differences in encryption also lead to differences in the decryption algorithm.

Compared to the decryption algorithm in UVW1, the UVW2 scheme introduces an additional verification step: checking whether the recovered error vector $\mathbf{e}'$ satisfies the condition $wt(\mathbf{e}') = w$. This is necessary because, in UVW2, the subset I_1 chosen during decryption may with non-negligible probability intersect the support of $\mathbf{e}_2$, i.e., $I_1 \cap \text{supp}(\mathbf{e}_2) \neq \emptyset$, which could lead to a decryption failure.

The following theorem 2 provides a theoretical bound on the probability of successful decryption for a single random choice of I_1 .

Theorem 2. *Let I_1 be the subset randomly chosen in Step 2 of the decryption algorithm in the UVW2 scheme. Then, the probability that $I_1 \cap \text{supp}(\mathbf{e}_2) = \emptyset$ is*

$$\mathbb{P}(I_1 \cap \text{supp}(\mathbf{e}_2) = \emptyset) = \frac{1}{\binom{n}{w}(q-1)^w} \left(\sum_{t=0}^{\frac{w}{2}} \sum_{i=0}^{w-2t} \binom{\frac{n}{2}}{t} \binom{\frac{n-2t}{2}}{i} \binom{\frac{n-2t-2i}{2}}{\frac{w-2t-i}{2}} \right. \\ \left. 2^i (q-2)^{\frac{w-2t-i}{2}} (q-1)^{\frac{w+i}{2}} \frac{\binom{\frac{n-w-i}{2}}{k_1}}{\binom{\frac{n-w+2t-i}{2}}{k_1}} \right).$$

Proof. Let $\mathbf{e} = (\mathbf{e}_1 || \mathbf{e}_2)$ be a vector of Hamming weight w , selected uniformly at random from the set $\mathcal{E}_{q,n}^w$, where $\mathbf{e}_1, \mathbf{e}_2 \in \mathbb{F}_q^{n/2}$. Define the following three intermediate weights:

$$\begin{aligned} |\mathbf{e}|^1 &= |\{i \in [n/2] \mid \mathbf{e}_j(i) = 0, \mathbf{e}_{3-j}(i) \neq 0, j = 1, 2\}|, \\ |\mathbf{e}|^2 &= |\{i \in [n/2] \mid \mathbf{e}_1(i) \neq 0, \mathbf{e}_2(i) \neq 0, \mathbf{e}_1(i) \neq \mathbf{e}_2(i)\}|, \\ |\mathbf{e}|^3 &= |\{i \in [n/2] \mid \mathbf{e}_1(i) \neq 0, \mathbf{e}_2(i) \neq 0, \mathbf{e}_1(i) = \mathbf{e}_2(i)\}|, \end{aligned}$$

Then the following identities hold:

$$|\mathbf{e}| = |\mathbf{e}|^1 + 2|\mathbf{e}|^2 + 2|\mathbf{e}|^3, \quad |\mathbf{e}_1 - \mathbf{e}_2| = |\mathbf{e}|^1 + |\mathbf{e}|^2.$$

We observe that when $|\mathbf{e}|^3 = 0$, it follows that $\text{supp}(\mathbf{e}_2) \subseteq \text{supp}(\mathbf{e}_1 - \mathbf{e}_2)$, and thus the event $I_1 \cap \text{supp}(\mathbf{e}_2) = \emptyset$ always occurs. However, when $|\mathbf{e}|^3 \neq 0$, both outcomes $I_1 \cap \text{supp}(\mathbf{e}_2) = \emptyset$ and $I_1 \cap \text{supp}(\mathbf{e}_2) \neq \emptyset$ are possible.

Based on the definitions of $|\mathbf{e}|^1$, $|\mathbf{e}|^2$ and $|\mathbf{e}|^3$, the probability that $I_1 \cap \text{supp}(\mathbf{e}_2) = \emptyset$ is given by:

$$\begin{aligned} \mathbb{P}(I_1 \cap \text{supp}(\mathbf{e}_2) = \emptyset) &= \sum_{t=0}^{\frac{w}{2}} \sum_{i=0}^{w-2t} \mathbb{P}(|\mathbf{e}|^3 = t, |\mathbf{e}|^1 = i, |\mathbf{e}|^2 = \frac{w-2t-i}{2}) \frac{\binom{\frac{n}{2}-i-\frac{w-2t-i}{2}-t}{k_1}}{\binom{\frac{n}{2}-i-\frac{w-2t-i}{2}}{k_1}} \\ &= \sum_{t=0}^{\frac{w}{2}} \sum_{i=0}^{w-2t} \mathbb{P}(|\mathbf{e}|^3 = t, |\mathbf{e}|^1 = i, |\mathbf{e}|^2 = \frac{w-2t-i}{2}) \frac{\binom{\frac{n-w-i}{2}}{k_1}}{\binom{\frac{n-w+2t-i}{2}}{k_1}} \\ &= \sum_{t=0}^{\frac{w}{2}} \sum_{i=0}^{w-2t} \left(\frac{1}{\binom{n}{w} (q-1)^w} \binom{n/2}{t} (q-1)^t \binom{n/2-t}{i} (2q-2)^i \right. \\ &\quad \left. \binom{n/2-t-i}{\frac{w-2t-i}{2}} ((q-2)(q-1))^{\frac{w-2t-i}{2}} \right) \frac{\binom{\frac{n-w-i}{2}}{k_1}}{\binom{\frac{n-w+2t-i}{2}}{k_1}} \\ &= \frac{1}{\binom{n}{w} (q-1)^w} \left(\sum_{t=0}^{\frac{w}{2}} \sum_{i=0}^{w-2t} \binom{n/2}{t} \binom{n-2t}{i} \binom{\frac{n-2t-2i}{2}}{\frac{w-2t-i}{2}} \right. \\ &\quad \left. 2^i (q-2)^{\frac{w-2t-i}{2}} (q-1)^{\frac{w+i}{2}} \frac{\binom{\frac{n-w-i}{2}}{k_1}}{\binom{\frac{n-w+2t-i}{2}}{k_1}} \right). \end{aligned}$$

This completes the proof of Theorem 2. $\square$

It is worth emphasizing that, for the parameter sets selected in Section 5, the probability that $I_1 \cap \text{supp}(\mathbf{e}_2) = \emptyset$ is very close to 1. For instance, under two parameter sets targeting 128-bit security, the probability is 0.9859 and 0.9781, respectively.

4 Security Analysis

In this section, we present the security proof for the public-key encryption scheme described in Section 3. Under the hardness assumptions of the General Decoding Problem (GDP) and the Code Distinguishing Problem (CDP), we prove that the UVW public-key encryption scheme (including both UVW1 and UVW2) achieves IND-CPA security.

4.1 IND-CPA Security

Since the UVW1 public-key encryption scheme is a simpler special case of the UVW2 scheme, we only prove the security of UVW1 here. We begin by describing the code distinguishing problem.

Let $\mathcal{G}_{\text{pk}}$ denote the set of all public-key matrices $\mathbf{G}_{\text{pk}}$ as described in Section 3.2. A code distinguisher $\mathcal{D}$ for $\mathcal{G}_{\text{pk}}$ is a probabilistic polynomial-time algorithm. Its distinguishing advantage is defined as:

$$\text{Adv}^{\mathcal{G}_{\text{pk}}}(\mathcal{D}) = \left| \mathbb{P}[\mathbf{G} \xleftarrow{\$} \mathcal{G}_{\text{pk}} \text{ code } [n, k] : \mathcal{D}(\mathbf{G}) = 1] - \mathbb{P}[\mathbf{G} \xleftarrow{\$} \text{ random } q\text{-ary code } [n, k] : \mathcal{D}(\mathbf{G}) = 1] \right|.$$

If the advantage $\text{Adv}^{\mathcal{G}_{\text{pk}}}(\mathcal{D})$ is negligible for any distinguisher $\mathcal{D}$, then the codes in $\mathcal{G}_{\text{pk}}$ are indistinguishable from random q -ary linear codes.

Definition 8. If for any distinguisher $\mathcal{D}$ with running time at most τ , it holds that

$$\text{Adv}^{\mathcal{G}_{\text{pk}}}(\mathcal{D}) \leq \epsilon.$$

then the $\mathcal{G}_{\text{pk}}$ code distinguishing problem is (τ, ϵ) -hard.

We now provide the security proof for the public key encryption scheme described in Section 3. Under the assumption that both the decoding problem and the distinguishing problem are hard, we prove that the scheme achieves IND-CPA security in the random oracle model.

Theorem 3. Suppose the $\mathcal{G}_{\text{pk}}$ code distinguishing problem and the general decoding problem are $(\tau_{CD}, \epsilon_{CD})$ -secure and $(\tau_{GD}, \epsilon_{GD})$ -secure, respectively. Then the public key encryption scheme described in Section 3 is IND-CPA secure in the random oracle model.

Proof. To prove that the encryption scheme satisfies IND-CPA security, we construct a sequence of games where each adjacent pair of games is computationally indistinguishable. Ultimately, we conclude that the ciphertexts generated from encrypting messages $\mathbf{m}_0$ and $\mathbf{m}_1$ are indistinguishable. If there exists an adversary $\mathcal{A}$ capable of distinguishing between any two neighboring games in this sequence, we can then use $\mathcal{A}$ to construct an algorithm $\mathcal{B}$ that either distinguishes the public key matrix from a random matrix or solves the general decoding problem.

Let **Game 0** (Figure 1) denote the real IND-CPA game in which the encryption scheme encrypts message $\mathbf{m}_0$. Denote by $\mathbb{P}(S_i)$ the probability that the adversary $\mathcal{A}$ outputs $b' = 0$ in **Game** i . Then we have:

$$\mathbb{P}(S_0) = \mathbb{P}[b' \leftarrow \mathcal{A}(\text{pk}, \lambda, (\mathbf{c}_1, \mathbf{c}_2)), b' = 0]. \quad (9)$$

In **Game 0**, the vector $\mathbf{r}$ is randomly chosen from the vector space $\mathbb{F}_q^k$, and the error vector $\mathbf{e} \in \mathcal{E}_{q,n}^w$ is randomly sampled from the set of length- n vectors with Hamming weight w and all nonzero entries being pairwise distinct.


```

1  $params \leftarrow \text{Setup}(1^\lambda)$ 
2  $(pk, sk) \leftarrow \text{KeyGen}(params)$ 
3  $(m_0, m_1) \leftarrow \mathcal{A}(pk, \lambda)$ 
4  $(c_1, c_2) = (rG_{pk} + e, m_0 \oplus \mathcal{H}(r, e)) \leftarrow \text{Enc}(pk, m_0)$ 
5  $b' \leftarrow \mathcal{A}(pk, \lambda, (c_1, c_2))$ 

```

Fig. 1 Game 0: The real IND-CPA game for encrypting the message m_0

```

1  $params \leftarrow \text{Setup}(1^\lambda)$ 
2  $(pk, sk) \leftarrow \text{KeyGen}(params)$ 
3  $pk \leftarrow G$ 
4  $(m_0, m_1) \leftarrow \mathcal{A}(pk, \lambda)$ 
5  $(c_1, c_2) = (rG + e, m_0 \oplus \mathcal{H}(r, e)) \leftarrow \text{Enc}(pk, m_0)$ 
6  $b' \leftarrow \mathcal{A}(pk, \lambda, (c_1, c_2))$ 

```

Fig. 2 Game 1

Game 1. In this game, the challenger replaces the public key matrix G_{pk} with a uniformly random matrix G (see Figure 2). To relate **Game 0** and **Game 1**, we construct a distinguisher $\mathcal{D}$. If the public key pk is generated via the KeyGen algorithm, then $\mathcal{D}$ essentially plays **Game 0**, and we have

$$\mathbb{P}[G_{pk} \leftarrow \text{KeyGen}(params) : \mathcal{D}(G_{pk}) = 1] = \mathbb{P}(S_0).$$

On the other hand, if the generator matrix in pk is randomly sampled, then $\mathcal{D}$ is playing **Game 1**, and we have

$$\mathbb{P}[G \xleftarrow{\$} \mathbb{F}_q^{k \times n} : \mathcal{D}(G) = 1] = \mathbb{P}(S_1).$$

Therefore, the advantage of $\mathcal{D}$ in distinguishing between the public key matrix G_{pk} and a uniformly random matrix G is given by

$$\text{Adv}^{CD}(\mathcal{D}) = |\mathbb{P}(S_0) - \mathbb{P}(S_1)|,$$

Since the code distinguishing problem for $\mathcal{G}_{pk}$ is $(\tau_{CD}, \epsilon_{CD})$ -secure, it follows that

$$|\mathbb{P}(S_0) - \mathbb{P}(S_1)| \leq \epsilon_{CD}. \quad (10)$$

```

1  $params \leftarrow \text{Setup}(1^\lambda)$ 
2  $(pk, sk) \leftarrow \text{KeyGen}(params)$ 
3  $(m_0, m_1) \leftarrow \mathcal{A}(pk, \lambda)$ 
4  $(c_1, c_2) = (rG + e, m_0 \oplus s) \leftarrow \text{Enc}(pk, m_0)$ 
5  $b' \leftarrow \mathcal{A}(pk, \lambda, (c_1, c_2))$ 

```

Fig. 3 Game 2

Game 2. In this game, the challenger replaces the hash output $\mathcal{H}(\mathbf{r}, \mathbf{e})$ with a uniformly random bitstring $\mathbf{s}$ (see Figure 3). We begin by analyzing the value of $\mathbb{P}(S_2)$ in **Game 2**. It is evident that the two components of the ciphertext, $\mathbf{c}_1$ and $\mathbf{c}_2$, are generated independently. Since $\mathbf{r}$, $\mathbf{G}$, and $\mathbf{e}$ are chosen independently of the message $\mathbf{m}_0$, $\mathbf{c}_1$ is independent of $\mathbf{m}_0$. Moreover, $\mathbf{c}_2$ is computed as $\mathbf{m}_0 \oplus \mathbf{s}$, where $\mathbf{s}$ is a random bitstring, making $\mathbf{c}_2$ statistically independent of $\mathbf{m}_0$ as well. As a result, the entire ciphertext $(\mathbf{c}_1, \mathbf{c}_2)$ reveals no information about the message $\mathbf{m}_0$, and the adversary can do no better than guessing b' at random:

$$\mathbb{P}(S_2) = \frac{1}{2}. \quad (11)$$

We now analyze the difference between **Game 1** and **Game 2**. In **Game 1**, if the adversary $\mathcal{A}$ successfully solves the general decoding problem and obtains the vectors $\mathbf{r}$ and $\mathbf{e}$, then $\mathcal{A}$ can correctly output b' . In contrast, in **Game 2**, even if the adversary $\mathcal{A}$ obtains $\mathbf{r}$ and $\mathbf{e}$ by solving the general decoding problem, $\mathcal{A}$ cannot learn anything about the b and can only guess b' at random. Therefore,

$$|\mathbb{P}(S_1) - \mathbb{P}(S_2)| \leq \epsilon_{GD}. \quad (12)$$

Combining equations (10), (11), and (12), we obtain

$$|\mathbb{P}(S_0) - \frac{1}{2}| \leq \epsilon_{GD} + \epsilon_{CD}. \quad (13)$$

Game 3. We now proceed with encrypting a different message, $\mathbf{m}_1$. Let **Game 3** (Figure 4) represent the real IND-CPA game in which the encryption scheme is applied to $\mathbf{m}_1$. As before, we denote by $\mathbb{P}(S_i)$ the probability that the adversary $\mathcal{A}$ outputs $b' = 0$ in **Game i** , so we have

$$\mathbb{P}(S_3) = \mathbb{P}[b' \leftarrow \mathcal{A}(\mathbf{pk}, \lambda, (\mathbf{c}_1, \mathbf{c}_2)), b' = 0]. \quad (14)$$

```

1  $params \leftarrow \text{Setup}(1^\lambda)$ 
2  $(\mathbf{pk}, \mathbf{sk}) \leftarrow \text{KeyGen}(params)$ 
3  $(\mathbf{m}_0, \mathbf{m}_1) \leftarrow \mathcal{A}(\mathbf{pk}, \lambda)$ 
4  $(\mathbf{c}_1, \mathbf{c}_2) = (\mathbf{rG}_{\mathbf{pk}} + \mathbf{e}, \mathbf{m}_1 \oplus \mathcal{H}(\mathbf{r}, \mathbf{e})) \leftarrow \text{Enc}(\mathbf{pk}, \mathbf{m}_1)$ 
5  $b' \leftarrow \mathcal{A}(\mathbf{pk}, \lambda, (\mathbf{c}_1, \mathbf{c}_2))$ 

```

Fig. 4 Game 3: The real IND-CPA game for encrypting the message $\mathbf{m}_1$

The only difference between **Game 3** and **Game 0** is that the message $\mathbf{m}_0$ has been replaced with $\mathbf{m}_1$.

Game 4. In this game, the challenger replaces the public key matrix $\mathbf{G}_{\mathbf{pk}}$ with a random matrix $\mathbf{G}$. The relationship between **Game 3** and **Game 4** is the same as

that between **Game 0** and **Game 1**, namely,

$$|\mathbb{P}(S_3) - \mathbb{P}(S_4)| \leq \epsilon_{CD}. \quad (15)$$

Game 5. In this game, the challenger replaces $\mathcal{H}(\mathbf{r}, \mathbf{e})$ with a uniformly random bit string $\mathbf{s}$. As in **Game 2**, the adversary has no advantage other than guessing b' randomly, hence

$$\mathbb{P}(S_5) = \frac{1}{2}. \quad (16)$$

Moreover, the relationship between **Game 4** and **Game 5** is the same as that between **Game 1** and **Game 2**, so we have

$$|\mathbb{P}(S_4) - \mathbb{P}(S_5)| \leq \epsilon_{GD}. \quad (17)$$

According to equations (15), (16) and (17), we get

$$|\mathbb{P}(S_3) - \frac{1}{2}| \leq \epsilon_{GD} + \epsilon_{CD}. \quad (18)$$

Finally, by combining (13) and (18), we conclude

$$|\mathbb{P}(S_0) - \mathbb{P}(S_3)| \leq 2(\epsilon_{GD} + \epsilon_{CD}), \quad (19)$$

which leads to the overall IND-CPA advantage bound:

$$\begin{aligned} Adv_{\mathcal{A}}^{\text{IND-CPA}} = & |\mathbb{P}[b' \leftarrow \mathcal{A}(\text{pk}, \lambda, \text{Enc}(\text{pk}, \mathbf{m}_0)), b' = 0] \\ & - \mathbb{P}[b' \leftarrow \mathcal{A}(\text{pk}, \lambda, \text{Enc}(\text{pk}, \mathbf{m}_1)), b' = 0]| \leq 2(\epsilon_{GD} + \epsilon_{CD}). \end{aligned} \quad (20)$$

□

4.2 The $(U + V, U + W)$ Code Distinguishing Problem

The security of the UVW1 scheme is based on two core hardness assumptions: 1. the decoding problem for random linear codes (Problem 1); 2. the problem of deciding whether a given linear code is a permutation of a $(U + V, U + W)$ code.

The first problem is known to be NP-complete, and even quantum algorithms require exponential time to solve it when parameters are chosen appropriately. The second hardness assumption implies that the public generator matrix $\mathbf{G}_{\text{pk}} = \mathbf{S}\mathbf{G}_{\text{sk}}\mathbf{P}$ is computationally indistinguishable from a random matrix.

From a structural perspective, a $(U + V, U + W)$ code is nearly indistinguishable from a random linear code of the same length and dimension, with only subtle differences appearing in the weight distribution of codewords—particularly among those with very low or very high Hamming weights. Specifically, when $\mathbf{v} = \mathbf{w} = \mathbf{0}$, the resulting codewords take the form $(\mathbf{u}, \mathbf{u})\mathbf{P}$; and when $\mathbf{u} = \mathbf{0}$, they take the form $(\mathbf{v}, \mathbf{w})\mathbf{P}$. These patterns may serve as potential indicators for distinguishing a structured code from a

truly random one. Consequently, the second assumption effectively reduces to identifying codewords with extreme weights a problem fundamentally equivalent in hardness to the first. Proposition 4 provides the weight distribution of codewords in an $[n, k]$ Reed-Solomon code over the finite field $\mathbb{F}_q$.

Proposition 4. *Let $\mathcal{C}$ be an $[n, k]$ Reed-Solomon code defined over the finite field $\mathbb{F}_q$. The sequence $(a_i)_{i=1}^k$ is defined as follows: set $a_1 = q - 1$, and for $i = 2, \dots, k$, define*

$$a_i = q^i - 1 - \sum_{j=1}^{i-1} \binom{n-k+i}{n-k+j} a_j$$

Let N_i denote the number of codewords in $\mathcal{C}$ with Hamming weight exactly $n - k + i$. Then, the number of such codewords is given by

$$N_i = \binom{n}{n-k+i} a_i.$$

Proof. Reed-Solomon codes are known to be maximum distance separable (MDS) codes. Consequently, for a given $[n, k]$ Reed-Solomon code $\mathcal{C}$, the Hamming weights of its codewords lie in the range $[n - k + 1; n]$. According to Definition 3, the dual code $\mathcal{C}^\perp$ of $\mathcal{C}$ is itself a Reed-Solomon code of parameters $[n, n - k]$. By the definition of a dual code, $\mathbf{x} \in \mathcal{C}$ if and only if

$$\mathbf{x}\mathbf{H}^\top = \mathbf{0}, \quad (21)$$

where $\mathbf{H}$ denotes a generator matrix of the dual code $\mathcal{C}^\perp$. Thus, computing the weight distribution of the code $\mathcal{C}$ is equivalent to analyzing the weight distribution of the solution space to the linear system defined by Equation (21).

Let $\mathbf{H}$ be a generator matrix of an $[n, n - k]$ Reed-Solomon code. By the definition of Reed-Solomon codes, any selection of $n - k$ columns from $\mathbf{H}$ forms a Vandermonde matrix, which is full-rank and thus has a non-zero determinant. This implies that these $n - k$ columns are linearly independent, and consequently, the linear system (21) admits no non-zero solution of Hamming weight $n - k$. On the other hand, any set of $n - k + 1$ columns from $\mathbf{H}$ must be linearly dependent, ensuring the existence of non-trivial solutions. This scenario can be viewed as a linear system with $n - k$ equations and $n - k + 1$ variables, yielding one degree of freedom. Therefore, the system (21) has solutions of Hamming weight exactly $n - k + 1$, and the number of such solutions is given by

$$\binom{n}{n-k+1} (q - 1).$$

We now analyze the number of solutions to the linear system (21) that have Hamming weight exactly $n - k + 2$. Select any $n - k + 2$ columns from the matrix $\mathbf{H}$, resulting in a system with $n - k$ linear equations and $n - k + 2$ variables. This system possesses two degrees of freedom, and hence the number of non-zero solutions is exactly

$$q^2 - 1.$$

By removing those solutions of weight $n - k + 1$, the number of solutions with weight exactly $n - k + 2$ for such a system is

$$q^2 - 1 - \binom{n - k + 2}{n - k + 1}(q - 1).$$

Therefore, the total number of solutions of weight $n - k + 2$ for the system (21) is

$$\binom{n}{n - k + 2} \left(q^2 - 1 - \binom{n - k + 2}{n - k + 1}(q - 1) \right).$$

Let

$$a_1 = q - 1, \quad a_2 = q^2 - 1 - \binom{n - k + 2}{n - k + 1}(q - 1).$$

We now compute the number of solutions of Hamming weight exactly $n - k + 3$ for the linear system (21).

As before, we consider choosing $n - k + 3$ columns from the matrix $\mathbf{H}$, forming a system with $n - k$ equations and $n - k + 3$ variables. This system has three degrees of freedom, and thus the number of non-zero solutions is

$$q^3 - 1.$$

To find the number of solutions with Hamming weight exactly $n - k + 3$, we subtract the number of solutions of weight $n - k + 1$ and $n - k + 2$, resulting in

$$\begin{aligned} q^3 - 1 - \binom{n - k + 3}{n - k + 1}(q - 1) - \binom{n - k + 3}{n - k + 2} \left(q^2 - 1 - \binom{n - k + 2}{n - k + 1}(q - 1) \right) \\ = q^3 - 1 - \sum_{j=1}^2 \binom{n - k + 3}{n - k + j} a_j. \end{aligned}$$

Let

$$a_3 = q^3 - 1 - \sum_{j=1}^2 \binom{n - k + 3}{n - k + j} a_j,$$

then the number of codewords in the solution space of the linear system (21) with Hamming weight exactly $n - k + 3$ is

$$\binom{n}{n - k + 3} a_3.$$

From the above analysis, we observe that selecting any $n - k + i$ columns from $\mathbf{H}$ forms a linear system with $n - k$ equations and $n - k + i$ variables. The number of solutions to this system with Hamming weight exactly $n - k + i$ equals the total number of non-zero solutions, $q^i - 1$, minus the number of solutions with weights $n - k + 1$

through $n - k + i - 1$. Hence, the number of solutions to (21) with weight $n - k + i$ is given by

$$\binom{n}{n - k + i} a_i,$$

where

$$a_i = q^i - 1 - \sum_{j=1}^{i-1} \binom{n - k + i}{n - k + j} a_j.$$

□

Having derived the weight distribution of codewords in a Reed-Solomon code, we now proceed to evaluate the statistical distance between the weight distribution of a Reed-Solomon code and that of a random linear code of the same length and dimension.

Proposition 5. *Let $\mathcal{C}_1$ be an $[n, k]$ Reed-Solomon code and $\mathcal{C}_2$ an $[n, k]$ random linear code over the finite field $\mathbb{F}_q$. Then, the statistical distance between the weight distributions of $\mathcal{C}_1$ and $\mathcal{C}_2$ is given by*

$$\rho(|\mathcal{C}_1|, |\mathcal{C}_2|) = \frac{1}{2} \left(\sum_{t=1}^{n-k} \frac{\binom{n}{t} (q-1)^t}{q^n} + \sum_{t=n-k+1}^n \left| \frac{\binom{n}{t} (q-1)^t}{q^n} - \frac{N_{t-n+k}}{q^k} \right| \right).$$

Proof. According to Proposition 4, the weight distribution of codewords in the Reed-Solomon code $\mathcal{C}_1$ can be characterized as follows:

1. For weights $t \in [1; n - k]$, the proportion of codewords with weight t is zero, i.e., such codewords do not exist in $\mathcal{C}_1$.
2. For weights $t \in [n - k + 1; n]$, the proportion of codewords with weight t is given by

$$\frac{N_{t-n+k}}{q^k}.$$

We now present the codeword weight distribution for a random $[n, k]$ linear code $\mathcal{C}_2$ over the finite field $\mathbb{F}_q$.

Since $\mathcal{C}_2$ is randomly selected from the ensemble of $[n, k]$ linear codes over $\mathbb{F}_q$, the expected proportion of codewords of weight t in $\mathcal{C}_2$ is given by

$$\frac{\binom{n}{t} (q-1)^t}{q^n}.$$

Based on the definition of statistical distance, the statistical distance between the weight distributions of $\mathcal{C}_1$ and $\mathcal{C}_2$ is

$$\rho(|\mathcal{C}_1|, |\mathcal{C}_2|) = \frac{1}{2} \left(\sum_{t=1}^{n-k} \frac{\binom{n}{t} (q-1)^t}{q^n} + \sum_{t=n-k+1}^n \left| \frac{\binom{n}{t} (q-1)^t}{q^n} - \frac{N_{t-n+k}}{q^k} \right| \right).$$

□

We evaluated the statistical distance under two parameter settings: for $(q, n, k) = (503, 500, 300)$, we obtain $\rho(|\mathcal{C}_1|, |\mathcal{C}_2|) \approx 2^{-1913.4}$; and for $(q, n, k) = (251, 250, 125)$, the distance is approximately $\rho(|\mathcal{C}_1|, |\mathcal{C}_2|) \approx 2^{-627.3}$. This indicates that, with appropriately chosen parameters, the weight distribution of a Reed-Solomon code is indistinguishable from that of a random code.

Moreover, this observation leads to the following corollary.

Corollary 1. *Let $\mathbf{G}_1$ be the generator matrix of an $[n, k]$ Reed-Solomon code $\mathcal{C}_1$ over the finite field $\mathbb{F}_q$. Let $\mathbf{G}_2$ and $\mathbf{G}_3$ be the generator matrices of two independent random $[n, k]$ linear codes $\mathcal{C}_2$ and $\mathcal{C}_3$ over $\mathbb{F}_q$, respectively. Define the linear code $\mathcal{C}'$ by the generator matrix $(\mathbf{G}_1 + \mathbf{G}_3 \| \mathbf{G}_3)$, and similarly define $\mathcal{C}''$ using $(\mathbf{G}_2 + \mathbf{G}_3 \| \mathbf{G}_3)$. Then the statistical distance between the weight distributions of $\mathcal{C}'$ and $\mathcal{C}''$ is equal to that between $\mathcal{C}_1$ and $\mathcal{C}_2$:*

$$\rho(|\mathcal{C}'|, |\mathcal{C}''|) = \rho(|\mathcal{C}_1|, |\mathcal{C}_2|) = \frac{1}{2} \left(\sum_{t=1}^{n-k} \frac{\binom{n}{t} (q-1)^t}{q^n} + \sum_{t=n-k+1}^n \left| \frac{\binom{n}{t} (q-1)^t}{q^n} - \frac{N_{t-n+k}}{q^k} \right| \right).$$

Since $\mathcal{C}''$ is a random linear code, it follows that the statistical distance between the weight distribution of $\mathcal{C}'$ and that of a random code with the same length and dimension is given by

$$\frac{1}{2} \left(\sum_{t=1}^{n-k} \frac{\binom{n}{t} (q-1)^t}{q^n} + \sum_{t=n-k+1}^n \left| \frac{\binom{n}{t} (q-1)^t}{q^n} - \frac{N_{t-n+k}}{q^k} \right| \right).$$

This implies that, under suitable parameter choices, the weight distribution of the $(U + V, U + W)$ code used in the UVW1 encryption scheme is indistinguishable from that of a code constructed from three random generator matrices, where the difference $\mathbf{G}_V - \mathbf{G}_W$ corresponds to the generator matrix of a Reed-Solomon code. As a result of this analysis, we conclude that recovering a codeword of the form $(\mathbf{v}, \mathbf{w})\mathbf{P}$ cannot be used as a distinguishing criterion to determine whether a given linear code is a permutation of a $(U + V, U + W)$ code.

We now analyze whether the presence of a codeword of the form $(\mathbf{u}, \mathbf{u})\mathbf{P}$ can serve as a distinguishing feature for identifying whether a given linear code is a permutation of a $(U + V, U + W)$ code. To this end, we treat the matrices $\mathbf{G}_U$, $\mathbf{G}_V$, and $\mathbf{G}_W$ as random and study the weight distribution of the code defined by them. The following proposition characterizes the expected weight distribution of a permuted $[n, k]$ $(U + V, U + W)$ code over the finite field $\mathbb{F}_q$.

Proposition 6. *Let $\mathcal{C}$ be a permuted $[n, k]$ $(U + V, U + W)$ code over $\mathbb{F}_q$, where U is a random $[\frac{n}{2}, k_1]$ linear code and V, W are both random $[\frac{n}{2}, k_2]$ linear codes. Let $N_{(\mathbf{u}, \mathbf{u})}$ and $N_{(\mathbf{u}+\mathbf{v}, \mathbf{u}+\mathbf{w})}$ denote the expected number of codewords of the form $(\mathbf{u}, \mathbf{u})\mathbf{P}$ and $(\mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{w})\mathbf{P}$, respectively. When w is even:*

$$N_{(\mathbf{u}, \mathbf{u})} = \frac{\binom{n/2}{w/2} (q-1)^{w/2}}{q^{n/2-k_1}}, N_{(\mathbf{u}+\mathbf{v}, \mathbf{u}+\mathbf{w})} = N_{(\mathbf{u}, \mathbf{u})} + \frac{\binom{n}{w} (q-1)^w}{q^{n-k}} \left(1 - \frac{1}{q^{k_2}}\right);$$

and when w is odd:

$$N_{(\mathbf{u}, \mathbf{u})} = 0, N_{(\mathbf{u}+\mathbf{v}, \mathbf{u}+\mathbf{w})} = \frac{\binom{n}{w}(q-1)^w}{q^{n-k}} \left(1 - \frac{1}{q^{k_2}}\right).$$

From the adversary's perspective, the primary strategy to distinguish a permuted $(U+V, U+W)$ code from a random linear code is to identify a codeword of the form $(\mathbf{u}, \mathbf{u})\mathbf{P}$ within the permuted $(U+V, U+W)$ code.

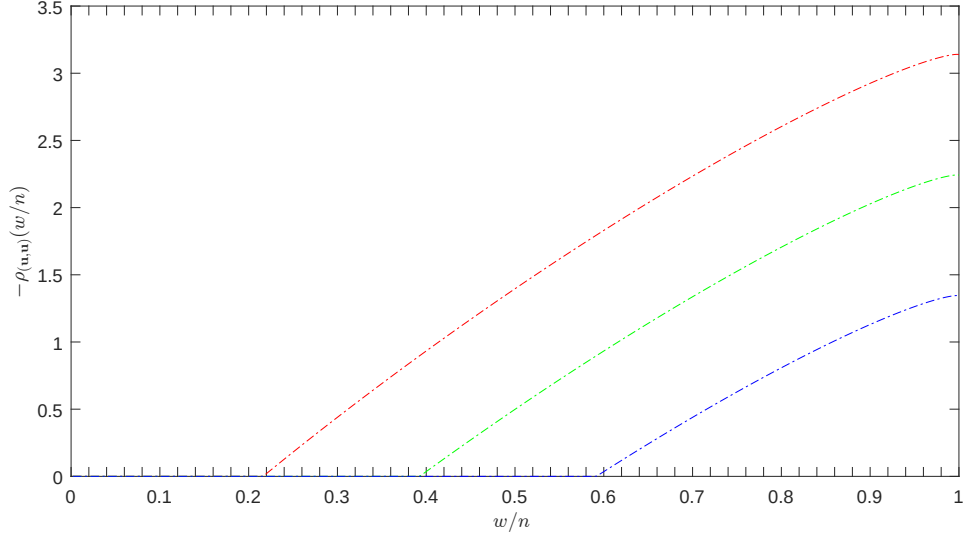


Fig. 5 Normalized log-density of codewords of the form $(\mathbf{u}, \mathbf{u})\mathbf{P}$ versus relative weight $\frac{w}{n}$

Figure 5 shows the normalized logarithmic density of codewords of the form $(\mathbf{u}, \mathbf{u})\mathbf{P}$ as a function of the relative weight $x = \frac{w}{n}$, defined by

$$\rho_{(\mathbf{u}, \mathbf{u})}\left(\frac{w}{n}\right) = \frac{\log_2(N_{(\mathbf{u}, \mathbf{u})}/N_{(\mathbf{u}+\mathbf{v}, \mathbf{u}+\mathbf{w})})}{n}.$$

All three curves in the figure are computed under the same parameters $(q, n, k) = (503, 1000, 500)$, and only differ in the values of k_1 and k_2 . From left to right, the curves correspond to $k_1 = 0.3k$, $0.5k$, and $0.7k$, respectively. We observe that when the relative weight w/n is less than 0.22, 0.4, and 0.59, respectively, almost all the codewords are of the form $(\mathbf{u}, \mathbf{u})\mathbf{P}$. Given the chosen parameters, finding a codeword with weight in the interval $[\frac{q-1}{q}(n-k)+1, k+\frac{q-1}{q}(n-k)] = [500.006, 999.006]$ is easy. As a result, when $k_1 = 0.7k$, an adversary can efficiently distinguish whether a given code is a permuted $(U+V, U+W)$ code. In contrast, for $k_1 = 0.3k$ or $0.5k$, such a distinction requires exponential time.

5 Parameter Selection and Comparison

5.1 Parameter Selection

As established earlier, the public key encryption scheme described in Section 3 achieves IND-CPA security under the assumption that both the code distinguishing problem and the general decoding problem are computationally hard. Since the distinguishing problem ultimately reduces to the problem of finding a codeword of a specific weight, we can rely on the known complexity of the information set decoding algorithm proposed by Niebuhr et al. [37] to provide a conservative security estimate for the scheme in Section 3. Taking the security parameter $\lambda = 128$ as an example, we present a parameter set that achieves the 128-bit security level.

It has also been shown that the dual of a $[n, k = k_1 + k_2]$ $(U + V, U + W)$ code $\mathcal{C}$ with parameters k_1, k_2 is itself a $[n, n - k]$ $(U + V, U + W)$ code with parameters $n/2 - k_2$ and $n/2 - k_1$, respectively. In particular, the dimension of the codewords in $\mathcal{C}$ of the form $(\mathbf{u}, \mathbf{u})\mathbf{P}$ is k_1 , while in its dual $\mathcal{C}^\perp$, the dimension of such codewords is $n/2 - k_2$. If $k_1 \neq n/2 - k_2$, then the parameter selection must account for the distinguishing complexity of both $\mathcal{C}$ and its dual $\mathcal{C}^\perp$. However, if $k_1 = n/2 - k_2$, then the distinguishing problems are equivalent for both codes. To simplify the analysis, we therefore select parameters such that $k_1 = n/2 - k_2$, allowing us to focus solely on the complexity of a single distinguishing problem.

As shown in Proposition 6, the number of codewords of the form $(\mathbf{u}, \mathbf{u})\mathbf{P}$ increases with the weight w , while their relative proportion decreases. In the context of information set decoding, a larger number of valid solutions generally leads to reduced computational complexity. Therefore, we define the following codeword weight parameter:

$$t_{max} = \max_w \left\{ w \mid \frac{\binom{n/2}{w/2} (q-1)^{w/2}}{q^{n/2-k_1}} > \frac{\binom{n}{w} (q-1)^w}{q^{n-k}} \left(1 - \frac{1}{q^{k_2}}\right) \right\}, \quad (22)$$

which is used as the target weight parameter in estimating the computational complexity of the distinguishing problem.

To ensure that codewords of weight t_{max} actually exist in the code $\mathcal{C}$, we define the minimum weight threshold as

$$t_{min} = \min_w \left\{ w \mid \frac{\binom{n/2}{w/2} (q-1)^{w/2}}{q^{n/2-k_1}} \geq 1 \right\}. \quad (23)$$

Consequently, if $t_{max} \geq t_{min}$, then the code $\mathcal{C}$ contains at least one codeword of weight t_{max} .

Define

$$\begin{aligned}
r_{t_{min}} &= \frac{\binom{n/2}{t_{min}/2} (q-1)^{t_{min}/2}}{q^{n/2-k_1}} \bigg/ \frac{\binom{n}{t_{min}} (q-1)^{t_{min}}}{q^{n-k}} \\
&= \frac{\binom{n/2}{t_{min}/2} q^{n/2-k_2}}{\binom{n}{t_{min}} (q-1)^{t_{min}/2}} \\
&\leq \frac{q^{n/2-k_2}}{\binom{n/2}{t_{min}/2} (q-1)^{t_{min}/2}}.
\end{aligned} \tag{24}$$

If $k_1 \leq k_2$, then

$$r_{t_{min}} \leq \frac{q^{n/2-k_2}}{\binom{n/2}{t_{min}/2} (q-1)^{t_{min}/2}} \leq \frac{q^{n/2-k_1}}{\binom{n/2}{t_{min}/2} (q-1)^{t_{min}/2}} \leq 1, \tag{25}$$

which implies that $t_{max} \leq t_{min}$. In other words, for any $t \geq t_{min}$, we have

$$\frac{\binom{n/2}{t/2} (q-1)^{t/2}}{q^{n/2-k_1}} \leq \frac{\binom{n}{t} (q-1)^t}{q^{n-k}} \left(1 - \frac{1}{q^{k_2}}\right). \tag{26}$$

This result indicates that distinguishing a $(U + V, U + W)$ code from a random code based on fixed-weight codeword search is infeasible. Therefore, the security evaluation can focus solely on the computational hardness of the decoding problem.

Example 1. To achieve 128-bit security, we choose parameters $(q, n, k, k_1, k_2) = (401, 800, 400, 200, 200)$. Let $w = 116 \leq n/2 - \sqrt{nk_2/2}$. Since $\binom{n}{w} (q-1)^w \ll q^{n-k}$ (a solution exists), we compute

$$\begin{aligned}
&WF_{qISD}(q, n, k, w) \\
&= \min_{l, v_1, v_2} \frac{1}{\sqrt{q-1}} \frac{\binom{n}{w} (q-1)^w}{\binom{n-k-l}{w-v} \binom{k}{v_1} \binom{l}{v_2} (q-1)^w} \left(K_q \frac{\binom{k}{v_1} \binom{l}{v_2} (q-1)^{v-1}}{q^l} \right. \\
&\quad \left. + \lambda_q^{-1} \left(\frac{2(q-1)l}{q \binom{l}{v'_2} (q-1)^{v'_2}} + v_2 \right) \times \sqrt{\binom{k}{v_1} \binom{l}{v_2} (q-1)^{v-1}} \right) \\
&\approx 2^{132.0113}.
\end{aligned} \tag{27}$$

where

$$\begin{aligned}
v &= v_1 + v_2, \quad v'_2 = \lfloor v_2/2 \rfloor, \\
\lambda_q &= 1 - \exp(-1) \approx 0.63, \quad K_q = 2w.
\end{aligned}$$

Table 1 provides the recommended parameter sets for the UVW public-key encryption scheme at various security levels when $k_1 = k_2$.

If $k_1 > k_2$, then $t_{max} > t_{min}$, and thus the computational complexity of both the decoding problem and finding fixed-weight codewords problem must be considered.

Table 1 Recommended Parameters for the UVW Public-Key Encryption Scheme
($k_1 = k_2$)

2^λ	q	n	k	w_0	w_1	w
2^{128}	401	800	400	100	16	116
2^{192}	601	1200	600	150	22	172
2^{256}	797	1592	796	199	31	230

Example 2. To achieve 128-bit security, we choose parameters $(q, n, k, k_1, k_2) = (337, 672, 336, 188, 148)$. Let $w = 112 = n/2 - \sqrt{nk_2/2}$. Since $\binom{n}{w}(q-1)^w \ll q^{n-k}$ (indicating that a solution exists), we compute

$$\begin{aligned}
& WF_{qISD}(q, n, k, w) \\
&= \min_{l, v_1, v_2, \sqrt{q-1}} \frac{1}{\sqrt{q-1}} \frac{\binom{n}{w}(q-1)^w}{\binom{n-k-l}{w-v} \binom{k}{v_1} \binom{l}{v_2} (q-1)^w} \left(K_q \frac{\binom{k}{v_1} \binom{l}{v_2} (q-1)^{v-1}}{q^l} \right. \\
&\quad \left. + \lambda_q^{-1} \left(\frac{2(q-1)l}{q \binom{l}{v'_2} (q-1)^{v'_2}} + v_2 \right) \times \sqrt{\binom{k}{v_1} \binom{l}{v_2} (q-1)^{v-1}} \right) \\
&\approx 2^{129.7357}.
\end{aligned} \tag{28}$$

where

$$\begin{aligned}
v &= v_1 + v_2, \quad v'_2 = \lfloor v_2/2 \rfloor, \\
\lambda_q &= 1 - \exp(-1) \approx 0.63, \quad K_q = 2w.
\end{aligned}$$

On the other hand, taking $t_{max} = 296$, we compute

$$\begin{aligned}
& WF_{qISD}(q, n, k, t_{max}) \\
&= \min_{l, v_1, v_2, \sqrt{q-1}} \frac{1}{\sqrt{q-1}} \frac{\binom{n}{t_{max}} q^{k_2}}{\binom{n-k-l}{t_{max}-v} \binom{k}{v_1} \binom{l}{v_2} \binom{n/2}{t_{max}/2} (q-1)^{t_{max}/2}} \left(K_q \frac{\binom{k}{v_1} \binom{l}{v_2} (q-1)^{v-1}}{q^l} \right. \\
&\quad \left. + \lambda_q^{-1} \left(\frac{2(q-1)l}{q \binom{l}{v'_2} (q-1)^{v'_2}} + v_2 \right) \times \sqrt{\binom{k}{v_1} \binom{l}{v_2} (q-1)^{v-1}} \right) \\
&\approx 2^{158.6948}.
\end{aligned} \tag{29}$$

Table 2 presents the recommended parameters for the UVW public-key encryption scheme under various security levels in the case where $k_1 > k_2$. Compared to Table 1, the parameter sets in Table 2 achieve the same security levels with smaller overall sizes.

5.2 Analyses and Comparisons

Table 2 Recommended Parameters for the UVW Public-Key Encryption Scheme ($k_1 > k_2$)

2^λ	q	n	k	k_1	k_2	w_0	w_1	w
2^{128}	337	672	336	188	148	94	18	112
2^{192}	499	996	498	279	219	139	28	167
2^{256}	677	1352	676	379	297	189	37	226

This subsection evaluates the key size of the UVW public-key encryption scheme and compares it with the Classic McEliece scheme, which has been submitted to NIST and advanced to the fourth round of evaluation.

In the UVW scheme, the public key is the generator matrix of the $(U + V, U + W)$ code. The matrix is originally of size $k \times n$, and can be transformed into systematic form with size $k \times (n - k)$. Since each matrix entry is an element of the finite field $\mathbb{F}_q$, the total size of the public key is $k \cdot (n - k) \cdot \lceil \log_2(q) \rceil$ bits.

Table 3 Public key sizes of the UVW schemes compared with the Classic McEliece schemes (Bytes)

2^λ	Scheme	(q, n, k, w)	pk
2^{128}	UVW	(401, 800, 400, 116)	180000
		(337, 672, 336, 112)	127008
	Classic McEliece	(2, 3488, 2720, 64)	261120
2^{192}	UVW	(601, 1200, 600, 172)	450000
		(499, 996, 498, 167)	279005
	Classic McEliece	(2, 4608, 3360, 96)	524160
2^{256}	UVW	(797, 1592, 796, 230)	792020
		(677, 1352, 676, 226)	571220
	Classic McEliece	(2, 6688, 5024, 128)	1044992

Table 3 presents a comparison of public key sizes between the UVW scheme and the Classic McEliece scheme. For security levels $\lambda = 128, 192, 256$, we adopt the recommended parameter sets of Classic McEliece, mceliece348864, mceliece460896, and mceliece6688128, as benchmarks. The results indicate that our proposed scheme offers significant advantages in terms of public key size, particularly when $k_1 > k_2$, where the public key size is approximately 50% that of the Classic McEliece scheme. Regarding private key management, as in other schemes, a seed can be used to deterministically generate the private matrix. In performance-critical scenarios, the matrix may be pre-computed and stored to accelerate decryption. In contrast, for applications with tight memory constraints and more relaxed latency requirements, only the seed needs to be stored, and the matrix can be generated during decryption. This allows flexible trade-offs between performance and storage

6 Conclusions

This paper proposes a code-based public-key encryption scheme, UVW, which can be regarded as a variant of the classic McEliece scheme. In our scheme, the $(U + V, U + W)$

code replaces the Goppa code used in McEliece, and the generator matrix of the Reed-Solomon code is split into two equally sized submatrices, which are embedded respectively into the generator matrix of the $(U + V, U + W)$ code. This improvement not only leverages the superior error-correcting capability of Reed-Solomon codes but also disrupts their vulnerable algebraic structure, thereby effectively resisting all known structure-based attacks. Our construction enhances the security of the scheme while allowing the use of shorter code lengths, leading to a significant reduction in public key size. Compared with the Classic McEliece scheme, which was submitted to the NIST post-quantum cryptography standardization process and successfully advanced to the fourth round, the UVW scheme demonstrates notable advantages in public key size.

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